ESSAYS AND STUDIES 2019

SLOW SCHOLARSHIP

Medieval Research and the Neoliberal University

Edited by CATHERINE E. KARKOV



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Slow Scholarship: Medieval Research and the Neoliberal University

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Catherine E. Karkov

for the English Association

D. S. BREWER

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Slow Digitisation and the Battle of the Books

ANDREW PRESCOTT

> *The battle facing the books today has nothing to do with arguments between Plato and Aristotle or Paganism and Christianity: it has to do with the survival of the books themselves.*
> Robin Alston

Among the colleagues during the twenty years I worked at the British Library who profoundly influenced me was the historian of the English language, bibliographer and librarian, Robin Alston (1933–2011).[1] Robin's great scholarly achievement was his twenty-volume *Bibliography of the English Language from the Invention of Printing to the Year 1800*,[2] but the range of his achievements and interests stretched far beyond this. While he was a lecturer at the University of Leeds, he founded Scolar Press to provide cheap facsimiles of historical and literary texts for his students. As editor-in-chief of the Eighteenth Century Short Title Catalogue, which afterwards became the English Short Title Catalogue or ESTC, Robin profoundly influenced the way in which we use printed books from the hand press period. Robin's insistence that this new catalogue should be machine readable and his energy in driving forward the ESTC laid the foundations of such current digital resources as Early English Books Online and Eighteenth-Century Collections Online. Robin was one of the pioneers who shaped the modern digital research environment for humanities scholars.

In 1990, Robin was appointed Professor of Library and Information Studies at University College London, and he gave his inaugural lecture at UCL, entitled 'The Battle of the Books', on 16 February 1993. I attended, with many others from the British Library.[3] The atmosphere was electric.

[1] Obituary by Stephen Green, The Guardian, 2 October 2011: https://www.theguardian.com/books/2011/oct/02/robin-alston-obituary (accessed 12 November 2018).

[2] R. Alston, *A Bibliography of the English Language from the Invention of Printing to the Year 1800: A systematic record of writings on English, and on other languages in English, based on the collections of the principal libraries of the world* (20 vols, Leeds: E.J. Arnold, 1965–2011).

[3] Sadly, this lecture was never published, but the text is available in a posting by Robin to the Humanist online discussion group, 7.176, 10 September 1993,

Robin was known as a charismatic, entertaining and thought-provoking lecturer who provided remarkable insights into current and future developments. He did not fail on this occasion. Robin took his starting point Jonathan Swift's satire, *A Full and True Account of the Battle Fought Last Friday between the Ancient and the Modern Books in St James's Library*, appended to the *Tale of a Tub* (1704). Swift imagined the books in the Royal Library joining in the conflict between those who revered classical learning and those who stressed the need for up-to-date modern learning. Swift described solitary volumes of classical learning being threatened by the massed ranks of thousands of modern books.

For Robin, Swift's satire encapsulated the dangers confronting librarianship as the web revolution was beginning. I remember that Robin's lecture was the first time I heard the words 'internet surfing', and many of the audience were astonished to learn that there were people who could earn a living from surfing databases. Robin was at the forefront of persuading librarians and bibliographers that they should exploit the potential of new forms of information technology, but on this occasion he sounded an alarm bell. Robin painted a picture of powerful commercial interests persuading libraries to invest heavily in computers that may not have been needed. He warned of the dangers of the creation of hypermarkets of information. 'The notion that knowledge can prosper by creating vast knowledge warehouses based on the hypermarket model – you can buy it if you can find it – is sheer fantasy as well as being intellectually suspect,' Robin declared. While emphasising how the availability of journal articles and new publications over networks could aid the cause of the moderns by making recent scholarship more accessible, Robin pointed out the risks of forgetting how the exploration of the works of the ancients in rare books and manuscripts was complex and time-consuming and that, for all the great electronic resources then becoming available, there were still vast quantities of printed and manuscript material inaccessible to the computer.

The contemporary battle of the books as described by Alston is a war between information and knowledge. It is a battle between moderns who propose that, by digitising every scrap of vellum and paper, all human understanding will become available, and ancients who argue that knowledge is gained by an extended intellectual journey in search of understanding about humanity. It is a battle between those who worship speed and efficiency and those who want to proceed more slowly and haphazardly. Robin described how science has improved the human condition, but pointed out how

http://lists.village.virginia.edu/lists_archive/Humanist/v07/0175.html (accessed 12 November 2018). All quotes from Robin's lecture are from this copy.

> the discoveries which have contributed to this have also had malign consequences: none more malign, I think, than the notion that knowledge is amenable to mechanical transference. Information without doubt, but knowledge decidedly not.

Robin graphically described the commercial and political forces that he believed were driving these processes:

> The temptation to transform research libraries into commercially driven surfing is already in evidence and conforms politically to the Friedman doctrine which holds that the notion of public service is obsolete and that libraries and education must be paid for by users or they will have to do without.

It was for these reasons that Robin argued that the modern battle of the books was not simply about forms of knowledge but was also a battle for the survival of the books themselves. For Robin, it was vital for librarians to develop new skills that blended ancient and modern techniques if they were to ensure that libraries were not replaced by new commercial networks and the superannuated books discarded.

Robin's lecture was remarkably prescient, and many of the dangers he foresaw twenty-five years ago are now urgent and daily threats. Robin's fear that powerful commercial interests would monopolise human culture by mass digitisation, perhaps abetted by librarians anxious to safeguard their own institutions, anticipated the appearance of Google Books and the way in which family history companies such as Ancestry and Findmypast have effectively privatised large swathes of historical records. Robin would have been horrified by the arbitrary way in which some librarians use the availability of digital surrogates to restrict access to original books and manuscripts, restricting the ability of scholars to investigate such physical features as watermarks or quiring structures. He would have been dismayed by the poor presentation in online packages such as Eighteenth-Century Collections Online of such features as the illustrations of eighteenth-century books. Robin also anticipated the rise of the MOOC in teaching and warned that digital access to course materials could be used as a means of avoiding the high costs of face-to-face teaching.

The modern battle of the books is between those moderns who see the value of books and manuscripts in their informational content and the ancients who prefer more slowly to explore the structure and character of books and manuscripts, using digital tools to investigate them more deeply. For the moderns, the important requirement is to undertake as much digitisation as quickly as possible; for the ancients, the value of

digitisation is in its ability to provide a variety of tools that facilitate highly detailed bibliographic analysis. The moderns cherish the ability of the computer to facilitate mass searching and distant reading; the ancients admire the way in which the computer enables them to go deeper and analyse with finer and finer degrees of granularity. Of course, it is not helpful to present these two approaches as completely antithetical. For example, the way in which the *Global Currents* project at Stanford and Groningen Universities automatically retrieves certain manuscript features such as *litterae notabiliores* and rubrics to facilitate comparison of scribal practice across a large number of twelfth-century manuscripts in the Parker Library at Corpus Christi College Cambridge may be seen at one level as using techniques of mass digitisation, but these methods are used to facilitate a detailed investigation of scribal behaviour, corresponding to the kind of bibliographic approach that Alston identified with the ancients.[4] Nevertheless, we can clearly discern a difference between those who would emphasise the managerial and efficiency gains of digitisation and those who would enlist digital tools in support of finer-grained and slower forms of analysis.

Robin Alston's 'Battle of the Books' was a warning against the dangers of mass digitisation. In a similar vein, in a recent article for *Archive Journal*, Lorna Hughes and I have argued for 'slow digitisation'.[5] We commented that:

> Too often, digitisation is treated as a form of colour microfilm, thereby offering distorted views of the manuscript and making it appear to be a simpler and more stable object than it really is. Digitisation provides a constantly expanding toolbox for probing and analysing manuscripts that goes beyond simple colour imaging. Like archaeological artefacts, manuscripts should be explored gradually, using a variety of technical aids and methods, building a multifaceted digital archive of the manuscript.

We defined 'slow digitisation' as follows:

> Just as there is a movement for 'slow food', so might we conceive of developing a movement advocating 'slow digitisation', in which rapid access is less important than the use of technological and other tools to

[4] https://globalcurrents.stanford.edu/ (accessed 20 November 2018).

[5] Andrew Prescott and Lorna Hughes, 'Why Do We Digitize? The case for slow digitization', *Archive Journal*, September 2018, http://www.archivejournal.net/essays/why-do-we-digitize-the-case-for-slow-digitization/ (accessed 12 November 2018).

> gradually excavate the complex layers that make up each manuscript. By 'slow digitisation' we mean not just the delivery of high-resolution digital images as a form of 'digital photocopying', where a single image capture of each page of a manuscript is presented as the definitive view of the folio as an object, but the use of advanced imaging techniques, including 3-D, RTI, and hyperspectral imaging – as well as specialised images such as those captured by raking light from angles other than a camera held directly above a flattened leaf. These two approaches – slow and mass digitisation – are by no means inimical, but they need to be held in balance.

In advocating for 'slow digitisation', Lorna and I were chiefly motivated by a concern that the full possibilities of digital imaging are not being exploited by libraries and archives, but there are also parallels between our call for 'slow digitisation' and the 'slow scholarship' movement discussed by Maggie Berg and Barbara Seeber in their 2016 publication, *The Slow Professor*.[6] Those who challenge the culture of speed in the academy see this as a route to restoring humanistic values to an educational system that has become degraded by managerial and commercial pressures. These concerns strongly echo those expressed by Robin Alston in 1993, who said that the growth of digital libraries may be 'a fine prospect for faculties of engineering, and the makers of electronic devices on which the librarian is increasingly dependent, but it might have serious consequences for a balanced view of who we are, where we came from, and where we are going'. Just as Alston called for librarians to develop a blend of ancient and modern outlooks, so likewise digital scholars need to be able to hold in balance slow digitisation and mass digitisation approaches.

Alston pointed out how much of our approach to digitisation has been shaped by the history of microfilm. The origins of microphotography go back to the nineteenth century, but it began to appear in commercial use in the 1920s.[7] The potential of microfilm for spreading knowledge appeared to be immense. In his book *World Brain* published in 1938, H.G. Wells noted that 'the British Museum library is making microfilms of the 4,000 books it possesses which were published before 1550',[8] and Wells anticipated that before long with the aid of microfilm 'any student, in any part of the world, will be able to sit with his projector in his own study at his or her convenience to examine *any* book, *any* docu-

[6] Maggie Berg and Barbara K. Seeber, *The Slow Professor: Challenging the culture of speed in the academy* (Toronto: University of Toronto Press, 2016).

[7] S. John Teague, *Microform, Video and Electronic Media Librarianship* (London: Butterworths, 1985), 8–9.

[8] H.G. Wells, *World Brain* (New York: Doubleday, 1938), 76–7.

ment, in an exact replica'.[9] Wells saw microfilm libraries as a key component in his vision of a world brain which, incorporating a constantly updated world encyclopaedia encapsulating human knowledge, was freely available to everyone. Wells hoped that the world brain, operated by an enhanced global educational system, would 'replace our multitude of unco-ordinated ganglia, our powerless miscellany of universities, research institutions, literatures with a purpose, national educational systems and the like'.[10]

The Library of Congress began to microfilm materials in British libraries in the early 1930s, and the British Museum became interested in the extent to which microfilm could be used to reduce wear and tear on fragile items in its collections.[11] In 1935, Eugene Power helped set up a programme at the British Museum for the microfilming of rare books. In 1938, Power established University Microfilms International at Ann Arbor in Michigan, and launched a programme to microfilm all the books published in English before 1700. During the Second World War, with funding from the Rockefeller Foundation and assistance from the American Council of Learned Societies, University Microfilms undertook a huge programme to microfilm manuscripts and archives in danger of loss and destruction in Britain. The microfilms produced by this British Manuscripts Project, which are still a valuable research resource, were catalogued at the University of Michigan Library, and sets of microfilm were deposited in the Library of Congress.[12]

The period after the Second World War saw the large-scale deployment of microform by libraries as a means of supporting preservation. Companies such as UMI (which became Bell and Howell in 1999 and ProQuest in 2001),[13] Gale (founded in 1954 and now part of the Cengage group),[14] Adam Matthew (now a subsidiary of Sage),[15] and Chadwyck-

9 Wells, *World Brain*, 77.

10 Wells, *World Brain*, xvi.

11 P.R. Harris, *A History of the British Museum Library, 1753–1973* (London: The British Library, 1998), 530–1; Teague, *Microform, Video and Electronic Media Librarianship*, 8–9.

12 'American Council of Learned Societies, British Manuscripts Project', *PMLA* 59, Supplement (1944), 1463–88; Lester K. Born, *British Manuscripts Project: A checklist of the microfilms prepared in England and Wales for the American Council of Learned Societies* (Washington, DC: Library of Congress, 1955).

13 https://www.proquest.com/about/history-milestones/ (accessed 20 November 2018).

14 https://en.wikipedia.org/wiki/Gale_(publisher) (accessed 20 November 2018).

15 https://en.wikipedia.org/wiki/Adam_Matthew_Digital (accessed 20 November

Healey (acquired by ProQuest in 1999),[16] produced short-run, high-cost microform publications of manuscript, archival and rare book materials that were intended for sale to a limited number of research libraries. These companies now dominate the commercial market for digital publication of primary materials. Libraries also developed collaborative projects for the microfilming of particular categories of material such as the Newsplan project for the systematic preservation of newspapers in Ireland and the United Kingdom.[17] There were also a handful of microfilming projects initiated and directed by academic communities, most notably the Anglo-Saxon Manuscripts in Microfiche Facsimile project (which now offers digital files instead of microfiche).[18]

Much of the way in which digital resources have developed is shaped by these initial experiences with microform. Many of the images in packages such as Early English Books Online, Eighteenth-Century Collections Online and the Burney Collection of Newspapers produced by former microform publishers such as Gale and ProQuest are scans from their microform publications. Even the images in some archival resources such as the online Ancient Petitions and probate copies of wills from the Prerogative Court of Canterbury produced by The National Archives are scans from microfilm. Early advocates of digital imaging such as Peter Robinson presented it chiefly as a more convenient and robust alternative to microfilm.[19] Microfilm offers the simplest imaginable view of a book or manuscript: a consecutive series of flat images offering only one fixed perspective on the volume. The reduction of manuscript pages to a single-size greyscale image strips the book of much of its distinctive identity as the book is reduced to transferable information. Looking at the original manuscript after seeing it on a microfilm reader can sometimes be a shock, as nothing in the microfilm has prepared you for the size and character of the original volume.

Given that so many of the digital resources we currently use are produced by firms that had their roots in microform publication, it is not surprising that microform precedents profoundly shape the way in which

2018).

16 http://www.dwyck.com/ (accessed 20 November 2018).

17 John Lauder, 'Partnerships in Preservation: The experience of the Newsplan 2000 Project', *IFLA Journal* 29:1 (2003), 4751.

18 Anglo-Saxon Manuscript in Microfiche Facsimile project (ASMMF), https://acmrs.org/publications/other/asmmf (accessed 12 November 2018).

19 Peter Robinson, *The Digitization of Primary Textual Materials* (Oxford: Office for Humanities Communication, 1993).

digital resources are presented. While there are, as we will see, a number of projects that offer alternative and more experimental views of how manuscripts and books can be digitally presented, for the most part the online presentation of manuscripts and other primary materials resembles colour microfilms. Much current digitisation effort has also been influenced by the example of Google Books, launched in 2002, which can in itself be seen as a resurrection of the vision of 1930s pioneers such as Wells and Power for the creation of universal global knowledge banks.[20] The launch of Google Books prompted a wave of anxiety among librarians as to how their institutions should position themselves in this new world. The way in which Google Books relied on partnerships with major research libraries brings to mind the concern expressed by Robin Alston that libraries might too easily sell out to commercial interests in pursuit of technological chimeras. The copyright difficulties and litigation encountered by Google Books, which led to a fundamental reshaping of the project, reflects another of the issues about creating knowledge hypermarkets to which Alston drew attention. Although Google Books offers the additional possibilities of keyword searching, nevertheless in many respects its vision of the structure and nature of the book is similar to that of early microfilm projects. European counterparts to Google Books, such as Europeana and Gallica,[21] although different in the way images are acquired and presented, likewise also seem to operate in the microfilm paradigm.

Digital images are easier to browse and manipulate than microfilm. We can readily magnify, rotate, crop and compare them. Moreover, colour digital images are cheaper and easier to produce than colour microfilm. We can more easily keep and share copies of digital images than microfilm. But otherwise the way in which digital images of primary textual materials are presented is often surprisingly similar to microfilm. The limitations will be familiar to anyone who regularly uses the digitised manuscripts collection on the British Library website. The current British Library manuscripts viewer (the new Universal Viewer is being rolled out for the British Library,[22] but it isn't clear how far it will improve matters) uses a Flash-based interface to restrict and control our interaction with the images. The result is very like an online microfilm viewer. You scroll

[20] The geopolitical and cultural implications of Google Books and Europeana are brilliantly analysed by Nanna Bonde Thylstrup in *The Politics of Mass Digitization* (Cambridge, MA: MIT Press, 2019).

[21] https://www.europeana.eu (accessed 12 November 2018); https://gallica.bnf.fr (accessed 12 November 2018].

[22] https://universalviewer.io/ (accessed 13 November 2018).

through the images sequentially or you can jump to a particular folio. You can view either a single page, facing pages, or the recto and verso of a folio. There are zoom controls. That's about it. The British Library and Bibliothèque Nationale de France have recently with the support of the Polonsky Foundation digitised 80 decorated manuscripts made in France and England between 700 and 1200, and a heavily curated website giving access to these manuscripts was launched in November 2018.[23] The Polonsky uses Mirador, one of the new generation of International Image Interoperability Framework (IIIF) viewers,[24] which offers more functionality in the way of comparing images from different repositories and undertaking more sophisticated image manipulation, but the precedent of the microfilm reader still looms large – the view options in Mirador offer a choice between a scroll view, which is like microfilm, a gallery view, which recalls microfiche and an image view, which combines both.

For Eugene Power and the early pioneers of microfilm, the aim was to undertake a one-off transfer of the information in books and manuscripts from one medium to another. Likewise, digitisation is currently often seen as a once and for all process. We talk of manuscripts being 'digitised' as if one set of shots is the beginning and end of the matter. However, we can be certain that it will before too long be necessary to take further digital images of the manuscripts covered by the Polonsky project. Camera quality will improve, and it will be possible to take higher-resolution images that show more detail. New techniques such as multi-spectral imaging or reflectance transformation imaging (RTI) offer the potential for fresh insights into the manuscript. We may want to use X-ray or CT imaging to examine under drawing in decorations. For the truly committed scholar seeking to understand the history of a manuscript or book, a single view will never be enough. A scholar exploring a manuscript may want to compare images of the manuscript made in 1940 with those from 1990 and 2010. Scholars wish to explore manuscripts from every different perspective. Digital images offer that possibility, if we use them imaginatively enough.

The pressures that drive mass digitisation are very similar to those underpinning the growth of the managerial audit culture in universities to which the slow scholarship movement is a reaction. Librarians and curators are encouraged to increase access and the number of people who engage with their collections. The number of items digitised and the number of

[23] https://manuscrits-france-angleterre.org/ (accessed 21 November 2018).
[24] https://projectmirador.org (accessed 12 November 2018).

hits received by each item are key performance indicators for many libraries, archives and museums. Documents like the British Library's annual report highlight the millions of pages digitised, and declare that 'every day, the Library's physical and digital spaces are teeming with people working to advance knowledge',[25] but it is unclear how the 25 million visits to the British Library's website during 2017–18 contributed to the advancement of knowledge.[26] Nevertheless, the Library's KPI targets for 2017–18 included such figures as 4 million website items consulted, 160,000 digital items acquired through legal deposit, and 6 million visitors to the British Library Learning website.[27] The emphasis throughout is on scale and large numbers, regardless of quality and with little regard for how such activities really advance knowledge.

In Britain, a pernicious expression of government and institutional pressure on libraries and archives to increase access through digitisation as quickly as possible has been the use of commercial partnerships to undertake digitisation projects, a cultural counterpart to the discredited Private Finance Initiative. The baleful effects of this are illustrated by the digitisation of newspapers. The Burney Collection of eighteenth-century newspapers held by the British Library, the largest and most important collection of eighteenth-century newspapers, was digitised by Gale Cengage.[28] The Burney Collection has been licensed by JISC Collections for use by UK universities, and is widely available to UK researchers. However, the second most important collection of early newspapers, the Nichols Collection in the Bodleian Library, was digitised by Adam Mathew, and this digital package is less widely available. Gale has recently produced its own version of the Nichols Collection, but this does not currently seem to have been acquired by many libraries. Indeed, if you want to consult the Nichols Collection digitally, the best thing to do is to go to the Bodleian Library, which seems to miss the whole point of digitisation.

This would be annoying enough, but in order to meet deadlines for the sale and disposal of its Newspaper Library at Colindale, the British Library arranged for the remainder of its newspapers to be digitised by

[25] *British Library Annual Report and Accounts 2017–18* (London: The British Library, 2018), HC 1255, 6.

[26] *British Library Annual Report 2017–18*, 37.

[27] *British Library Annual Report 2017–18*, 36–7.

[28] Andrew Prescott, 'Searching for Dr Johnson: The digitisation of the Burney Newspaper Collection', in *Travelling Chronicles: News and newspapers from the early modern period to the eighteenth century*, ed. S.G. Brandtzæg, P. Goring and C. Watson (Leiden: Brill, 2018), 49–71.

the family history firm Findmypast. As a result, access to the British Newspaper Archive requires an individual subscription of at least £85 a year (depending on which package you choose).[29] Library subscriptions do not appear to be available, and few university libraries offer access to the British Newspaper Archive. There is free access if you go to the British Library itself, but again this seems to run counter to the very concept of digitisation enhancing access. Two decades of extensive effort and investment in digitisation has indeed enhanced the availability of newspapers, but the haste to digitise has produced a fragmented and confusing landscape. Curiously, the microfilm approach to newspapers was far more structured. The microfilm edition of eighteenth-century newspapers attempted to provide integrated runs of titles from the Burney and Nichols collections,[30] while Newsplan achieved systematic microfilm of nineteenth-century material. The fragmented digital coverage of UK newspapers recalls the words of Robin Alston in 1993:

> It seems to me an irony that would not have escaped Swift that at precisely the point in time when scholarship has begun to accept the principle of the unity of knowledge and the value of interdisciplinary research we are bent upon its fragmentation.

The pressure to maximise digital coverage has led many archives to rely on family history firms to digitise as much material as possible, with little regard for how the digital record is then made available. The most popular activity on the web after pornography is family history research,[31] and there is big money involved. The British Library holds the India Office Records, which include many of the genealogical records for India during British rule. In order to get these records digitised as quickly as possible, the British Library struck a deal with the family history website Findmypast.[32] Findmypast digitised the records, but if you want to

[29] https://www.britishnewspaperarchive.co.uk/account/subscribe (accessed 13 November 2018).

[30] http://microformguides.gale.com/Data/Introductions/10030FM.htm (accessed 20 November 2018).

[31] Bruce Falconer, 'Ancestry.com's Genealogical Juggernaut', *Businessweek*, 20 September 2012, http://www.businessweek.com/articles/2012-09-20/ancestry-dot-coms-genealogical-juggernaut (accessed 13 February 2014); Gregory Rodriguez, 'How Genealogy Became Almost as Popular as Porn', *Time*, 30 May 2014, http://time.com/133811/how-genealogy-became-almost-as-popular-as-porn/ (accessed 12 November 2018).

[32] 'Hidden History of the British in India', British Library press release, 29

consult the records outside the British Library, you need a subscription to Findmypast, which can cost as much as £156 a year.[33] These records are not generally available in university libraries. Under pressure to digitise as much as possible, many archives have made agreements allowing firms like Ancestry and Findmypast to digitise key records such as the census, parish registers and rate books. In order to access these records you need subscriptions to these companies. You also have to grapple with interfaces that are aimed at amateur family historians and are unsuited for complex queries. This is nothing less than the privatisation by stealth of our historical heritage in order to fulfil arbitrary management targets to demonstrate increased use and access.

We need alternative models of digital engagement. We need urgently to think beyond the commercial microform precedent in shaping our digital resources. I cut my digital teeth working with Kevin Kiernan on the *Electronic Beowulf* project in the 1990s. *Electronic Beowulf* was one of the first digital editions produced under the auspices of the British Library and includes a complete colour facsimile of Cotton MS Vitellius A.xv, which contains the only known medieval text of *Beowulf*.[34] The British Library has since digitised Cotton MS Vitellius A.xv anew at a much higher resolution, and these images are available on the British Library website.[35] It might therefore be assumed that *Electronic Beowulf* has been superseded by the new digital facsimile on the British Library site, but this is far from the case. As a result of the fire damage to the manuscript in 1731, many readings in the manuscript can only be made out under ultra-violet light. This has been known for many years, and in the black-and-white facsimile of the Nowell Codex edited by Kemp Malone and published in *Early English Manuscripts in Facsimile* in 1963,[36] the British Museum photographer reproduced some pages

January 2014, https://www.bl.uk/press-releases/2014/january/hidden-history-of-the-british-in-india (accessed 14 November 2018).

33 Rosemary Collins, 'Findmypast introduces new pricing structure', *Who Do You Think You Are?*, 27 November 2017, http://www.whodoyouthinkyouaremagazine.com/news/findmypast-introduces-new-pricing-structure (accessed 14 November 2018).

34 http://ebeowulf.uky.edu (accessed 13 November 2018). For information about the history of the project, see http://www.uky.edu/~kiernan/eBeo_archives/ (accessed 14 November 2018).

35 http://www.bl.uk/manuscripts/FullDisplay.aspx?index=4&ref=Cotton_MS_Vitellius_A_XV (accessed 13 November 2018).

36 Kemp Malone (ed.), *The Nowell Codex: British Museum Cotton Vitellius A. XV, second MS.* Early English Manuscripts in Facsimile 12 (Copenhagen: Rosenkilde

under ultra-violet light. The digital version on the British Library's digitised manuscripts website does not include any imaging under ultra-violet light. As a result, readings that are visible in the black-and-white photographic facsimile cannot be seen in the colour digital images on the British Library's own website. On the other hand, the imaging of readings under special lighting conditions was a particular focus of *Electronic Beowulf*, which provides comprehensive pictorial coverage of such readings. In short, anyone engaged in serious work on the *Beowulf* manuscript should always use the older *Electronic Beowulf* rather than the more recent defective British Library version.

The fact that the British Library's library of digitised manuscripts can include a facsimile of the Nowell Codex that is less useful for research purposes than the 1963 black-and-white facsimile illustrates the dangers of a one-size-fits-all production-line approach to the digitisation of manuscripts. By contrast, *Electronic Beowulf* epitomises the virtues and potential of a slow digitisation approach. This is because it was driven by the scholarly vision and profound knowledge of the manuscript of its editor Kevin Kiernan, rather than deriving from a more mechanistic digitisation programme. *Electronic Beowulf* provides high-resolution colour images of Cotton MS Vitellius A.xv in its entirety, but it goes much further than this. All readings under ultra-violet light are imaged and documented. Moreover, the hundreds of letters in the manuscript concealed under paper frames when the manuscript was conserved in the nineteenth century were imaged using fibre-optic backlighting, something that had not been possible until digital cameras were available. In this way, a mass of new information about the *Beowulf* text was recovered and documented.

This intensive forensic imaging of Cotton MS Vitellius A.xv is accompanied in *Electronic Beowulf* by an extensive image archive of primary materials relating to the transmission of the *Beowulf* text. Users can compare readings in the manuscript with both sets of transcripts made in the eighteenth century for the Danish antiquary Thorkelin, and with the detailed collations of Thorkelin's edition made by Sir Frederic Madden and John Conybeare, offering a comprehensive archive of information about readings in the manuscript prior to its conservation in 1845. *Electronic Beowulf* is far from being the sort of do-it-yourself edition that Robin Alston feared, however. It is accompanied by a transcript and full edition by Kevin Kiernan that incorporates sophisticated

& Bagger, 1963).

searching and options allowing metrical, grammatical and other aspects of the text to be readily explored.

In the way in which *Electronic Beowulf* uses imaging to open up the various layers of evidence within the manuscript and facilitates comparison with other relevant materials, *Electronic Beowulf* fulfils the vision of slow digitisation that Lorna Hughes and I outlined in our *Archive Journal* article:

> 'Slow digitisation' would gradually explore the different layers of evidence in a manuscript, just as an archaeologist might very slowly and carefully examine a pot. Rather than creating one single digital representation of a manuscript to support 'access,' the use of a variety of digital tools to gradually explore a manuscript in-depth can generate an archive of information about one single manuscript that, as well as supporting scholarship, also provides a much deeper educational experience.[37]

Whereas the simplistic representation of Cotton MS Vitellius A.xv on the British Library's digitised manuscript site conceals vital evidence about the manuscript and misleads the viewer into thinking the *Beowulf* text is simpler than it is, *Electronic Beowulf* confronts the user at every point with the complexity and difficulty of the manuscript evidence for *Beowulf*. Reminding us how complex, deceptive and bewildering manuscripts can be should surely be one of the functions of digitisation. But it isn't a process that can be done simply and quickly. Creating elaborate digital archives like *Electronic Beowulf* is an expensive, time-consuming and above all slow business. It is slower and more intellectually taxing than writing articles and books. It requires a large team of collaborators and many different resources. It won't fit easily in research assessment timetables. But ultimately it will help generate more profound knowledge about the manuscript than a series of quick snapshots. This is slow scholarship, on the side of the ancients, but it is surely what humanities scholars should be doing.

It may seem that the *Electronic Beowulf* is an unusual example, because the damage to the manuscript in 1731 means that use of special lighting and other techniques is essential in imaging the manuscript. But of course many other Cotton manuscripts were also damaged in 1731, and the use of the approaches pioneered in *Electronic Beowulf* will yield dividends in investigating these manuscripts as well. A.S.G. Edwards has recently criticised the British Library for failing to use digital methods to examine the

[37] Prescott and Hughes, 'Why Do We Digitise'?

burnt Cotton manuscripts.[38] Edwards doesn't mention *Electronic Beowulf*, but nevertheless the potential of these methods to further the process of exploring the burnt Cotton manuscripts is immense. It is consequently baffling that the British Library, in pursuit of fulfilling KPIs by maximising the number of items digitised, persists in adding to its site images of manuscripts badly damaged in 1731 that cry out for the kind of special treatment developed by *Electronic Beowulf* project. Kevin Kiernan demonstrated in the *Electronic Boethius* project how much can be recovered from the Alfredian translation of the *Consolation of Philosophy* in Cotton MS. Otho A.vi by the use of ultra-violet light and imaging processing.[39] This makes it bizarre that the British Library only shows plain light images of Otho A.vi on its digitised images website. Similarly, the Gildas manuscript, Cotton MS. Vitellius A.vi, has many badly damaged folios that would benefit from multi-spectral imaging, but again the British Library only offers plain light images. The use of backlighting to reveal sections of the manuscript concealed by paper frames is also a technique that has enormous potential to recover letters and words concealed since the nineteenth century in many Cotton manuscripts, but there is no sign of the British Library experimenting with such techniques. Although the British Library has developed great expertise in multi-spectral imaging, there does not appear to be any systematic programme for its use. Multi-spectral imaging of very badly burnt manuscripts such as Cotton MS Otho B.x, an early eleventh-century collection of mostly saints' lives by the monk Ælfric (but currently undescribed in the British Library catalogue), has enormous potential for recovery of hitherto unstudied texts, as Kevin Kiernan demonstrated twenty-five years ago.[40]

There are many other exemplars of slow digitisation. Bill Endres's remarkable work on the St Chad Gospels not only provides spellbinding

[38] A.S.G. Edwards, 'The Digital Archive, Scholarly Enquiry, and the Study of Medieval English Manuscripts', *Archive Journal*, September 2018, http://www.archivejournal.net/essays/digital-archive-scholarly-enquiry-and-the-study-of-medieval-english-manuscripts/ (accessed 15 November 2018).

[39] Kevin Kiernan, 'Alfred the Great's Burnt Boethius', in *The Iconic Page in Manuscript, Print and Digital Culture*, ed. G. Bornstein and T. Tinkle (Ann Arbor, MI: University of Michigan Press: 1998), 7–32; Kevin Kiernan et al., 'The ARCHway Project: Architecture for Research in Computing for Humanities through research, teaching, and learning', *Literary and Linguistic Computing* 20 (2005), 69–88; Susan Irvine, 'Fragments of Boethius: The reconstruction of the Cotton manuscript of the Alfredian text', *Anglo-Saxon England* 34 (2005), 169–81.

[40] Kevin Kiernan, 'British Library MS Cotton Otho B. x, fol. 13(54v)r', http://www.uky.edu/~kiernan/eBeo_archives/OBx/ (accessed 13 November 2018).

images of a beautiful insular manuscript, but the range of imaging techniques used by Bill provides many varied perspectives on the manuscript.[41] The 3D images of the Chad Gospels can help anticipate where conservation issues might arise in the manuscript. Bill has developed an archive of images of the Chad Gospels that allow issues like pigment deterioration to be pinpointed. The Chad Gospels project provides comprehensive coverage of views of the manuscript under different light wavelengths. While the British Library seems more concerned in its digitised manuscripts website with maximising coverage, it has nevertheless hosted some good examples of slow digitisation methods, such as its digital version of the early Biblical manuscript, Codex Sinaiticus.[42] The digital Codex Sinaiticus reunites parts of the manuscript currently split between London, St Petersburg, Leipzig and Mount Sinai. As well as images under standard light, it also provides images under raking light, enabling ruling, erasures and other information to be easily made out. The images are accompanied by a transcription in Greek and English translation. The digital Codex Sinaiticus invites us to explore the complexities of the manuscript rather than providing a flat single-dimensional view of it.

Slow digitisation is not necessarily restricted to projects using special lighting and imaging techniques. Another standard bearer for slow digitisation at the British Library is the International Dunhuang Project.[43] The Buddhist cave library near the city of Dunhuang on the edge of the Gobi Desert contained approximately 40,000 manuscripts, paintings and printed documents documenting the rich culture of life on the Silk Road between 100 BCE and 1000 CE. This cave was sealed up at the end of the first millennium and only rediscovered by archaeologists in 1900. The contents of the cave library, however, were dispersed across many institutions throughout the world, making it difficult to understand the nature and context of the library. The International Dunhuang Project is a vast undertaking that seeks to digitally reintegrate the Dunhuang cave library and add to it tens of thousands of other artefacts discovered by archaeologists in Silk Road cities at the beginning of the twentieth century. The International Dunhuang Project is a large-scale digitisation project embracing many international partners whose database currently contains over 522,000 images. But the project is driven by the vision of slowly and meticulously reuniting these huge dispersed collections. It is motivated by the scholarly need to understand the context and

41 https://lichfield.ou.edu/ (accessed 19 November 2018).

42 https://www.codexsinaiticus.org (accessed 19 November 2018).

43 https://idp.bl.uk (accessed 19 November 2018).

structure of these collections. It has over a period of twenty-five years sought to enhance our understanding of these documents in ways that would not otherwise be possible. In these ways, it epitomises the values of slow digitisation.

Slow digitisation is just as relevant to modern records as to ancient and medieval manuscripts. When David Livingstone was stuck in the Congolese village of Bambarre for nearly seven months between 1870 and 1871, he ran low on supplies, and ran out of ink and paper. Using scraps of a newspaper and old letters, however, Livingstone was able to keep writing his field diary using local African dye made from plants. These manuscripts were deposited in the National Library of Scotland, but the dye-ink faded and the diaries could not be read with the naked eye. However, the David Livingstone Spectral Imaging project, directed by Professor Adrian Wisnicki of Indiana University of Pennsylvania and Birkbeck, University of London, used multi-spectral imaging to retrieve Livingstone's text and prepare an edition of his 1871 field diary.[44]

An illustration of the potential of slow digitisation approaches for early modern records is the *Signed, Sealed and Undelivered* project involving collaborators from MIT, Utrecht University, Leiden University, University of Groningen, University of Oxford, Queen Mary University of London and King's College London.[45] The postmasters of the Dutch city of The Hague at the end of the seventeenth century, Simon de Brienne and his wife Maria Germain, were at the centre of European communication networks at the time when William of Orange was seizing the British throne. The Briennes left behind a trunk containing 2,600 unopened letters. Using the latest advances in X-ray imaging from the field of dentistry developed by David Mills and the Apocalypto group at Queen Mary University of London, the project is reading and editing these sealed letters without opening them, which would destroy important evidence on letter-writing practice.

Manuscripts and books of all types and periods constantly reveal new secrets and surprises when interrogated in this way. A slow digitisation approach assumes there is always something fresh to discover in a manuscript, no matter how well known it is and how often it has been imaged. A good illustration of this is the hyperspectral imaging by the Library of Congress of Thomas Jefferson's draft of the Declaration of Independence, which revealed how Jefferson changed the word 'subjects' to 'citizens'.[46]

[44] https://livingstone.library.ucla.edu (accessed 19 November 2018).

[45] https://brienne.org.

[46] 'Hyperspectral Imaging by Library of Congress Reveals Change Made by

Multi-spectral imaging has also been used recently to study the writing process of Jorge Luis Borges.[47]

But slow digitisation is expensive, time-consuming and demanding in resources. *Electronic Beowulf* required the collaboration of academics, conservators, curators, photographers and computer scientists. In many ways one of the most exciting outcomes of *Electronic Beowulf* was the way in which it enabled these different groups to discuss and debate the digitisation process rather than (as is too often the case now) simply accept what large institutions like the British Library are willing to provide. In the early days of digitisation, it seemed that one of the great potential benefits of digitisation would be that collaborative projects like *Electronic Beowulf* would promote greater dialogue and shared working between libraries and archives and academic researchers. It was hoped that this might create new disciplinary and research configurations. However, this has not happened to the extent that might have been anticipated. Libraries have happily reverted to offering 'users' set patterns of service and digital provision on something of a take-it-or-leave-it basis. Academics are often puzzled as to how to build a dialogue. Managerially, of course, this is a much less demanding approach, but does it promote the shared creation of knowledge? We might hope that 'slow digitisation' might break down such barriers.

It is understandable that librarians, under political pressure to increase use of and access to their collections, jib at the cost of what has been called 'boutique digitisation', but this is perhaps partly because of ways in which the arguments for digitisation have been expressed. Major libraries take great pride in the slow and costly processes by which rare books and manuscripts are conserved, and see this as part of a necessary investment in our shared cultural heritage. Perhaps we should also view the slow and meticulous digitisation of manuscripts like *Beowulf*, which add immeasurably to our understanding of these cultural monuments, in a similar way.

The contemporary battle of the books is part of a war of resistance against audit, efficiency, key performance indicators, management lifecycles and workload models. It is a battle to protect the books from being

Thomas Jefferson in Original Declaration of Independence Draft', Library of Congress press release, 2 July 2010, https://www.loc.gov/item/prn-10-161/analysis-reveals-changes-in-declaration-of-independence/2010-07-02/ (accessed 19 November 2010).

47 Nora Benedict, 'Digital Approaches to the Archive: Multispectral imaging and the recovery of Borges's writing process in "El muerto" and "La casa de Asterión"', *Variaciones Borges* 45 (2018), 153–69.

overwhelmed by the amorphous idea of information. Our battle resists information. At the heart of computational thinking is Claude Shannon's information theory, which sees a distinction between the message, comprising data consisting of bits, and the medium carrying the message. This idea is embedded in much editorial theory in the humanities and is fundamental to such key aspects of digital humanities as the Text Encoding Initiative. Yet we cannot disentangle the history of a poem such as *Beowulf* from the manuscript in which it is preserved. Our understanding of a complex poem such as *Piers Plowman* depends fundamentally on our understanding of the many manuscripts in which it is preserved. For the humanities, the medium and the information are profoundly and inescapably intertwined. Much of what we study in the humanities is the history and result of that interaction. By assuming that the information in a book can be separated from its physical context and transferred to a new medium, we risk losing all understanding of the nature of the book. Books and manuscripts are not information.

Slow digitisation resists assumptions about the priority of information and seeks to enlist digital tools in a more dynamic and pro-active engagement with books and manuscripts as artefacts. Of course, there is a place for mass digitisation approaches, but we should not assume that this is the only approach. 'One size fits all' is never a helpful approach in humanities scholarship. Beside the mass digitisation programmes, it is vital that there is also space for the more experimental probing and investigation that is at the heart of slow digitisation. Books and manuscripts are after all craft productions, and by likewise using digital methods in a craft manner, rather than using production- line methods, we will enhance our understanding of the way in which books and manuscripts were created. By using craft methods of digitisation, we will develop a fuller and more rounded understanding of the objects we digitise. The battle of the books is one that defends a view of books, manuscripts and archives as complex artefacts not as carriers of information. Slow digitisation is an ideology of resistance against superficial views of information and, as Robin Alston first reminded us, at stake in this battle is the survival of the books themselves.